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Question 1: Algorithm Design
and Asymptotic Analysis

1. Strategy:

We use a linear Scan Strategy. We set

minVal and $\text{maxVal} \leftarrow A[0]$, then iterate

$A[1 \dots n-1]$ to compare and update minVal and maxVal .

2. pseudocode:

Algorithm FindMinMax (A, n)

$\text{minVal} \leftarrow A[0]$

$\text{maxVal} \leftarrow A[0]$

for $i \leftarrow 1$ to $n-1$ do

if $A[i] < \text{minVal}$ then $\text{minVal} \leftarrow A[i]$

elseif $A[i] > \text{maxVal}$ then $\text{maxVal} \leftarrow A[i]$

endfor

return ($\text{minVal}, \text{maxVal}$)

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Design
Analysis

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3- primitive operations.

Assignments: $\text{minVal} \leftarrow A[0], \text{maxVal} \leftarrow A[0], i \leftarrow i + 1$.

eg. Wreset

Array indexing: $A[0], A[i]$.

then iterate

Comparison: $A[i], A[i]$, $A[i] > \text{maxVal}, i \leftarrow i$.

and update

4- Best-case and worst:

Best-case: Array is descending order (e.g., $[5, 4, 3, 2, 1]$).

Worst-case: Array is ascending order (e.g., $[1, 2, 3, 4, 5]$).

5- Asymptotic Notation:

Val $\leftarrow A[i]$

Best-case: $O(n)$.

maxVal $\leftarrow A[i]$

Worst-case: $O(n)$.

Overall Time Complexity: $O(n)$.

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6 - Space Complexity:

Space Complexity: $O(1)$ Auxiliary

Space uses only minVal, maxVal.

Question 2: Design and compare

1 - pseudocode:

Algorithm 1: Linear Search (A, n)
for $i \leftarrow 0$ to $n-1$ do
 if $A[i] = x$ then return i
end for
return -1.

Algorithm 2:

Algorithm Binary Search (A, n, x)

low $\leftarrow 0$, high $\leftarrow n-1$

while low $\leq$ high do

 mid $\leftarrow$ low + (high - low) / 2

 if $A[mid] = x$ then return mid

 else if $A[mid] < x$ then low $\leftarrow$ mid + 1

 else high $\leftarrow$ mid - 1

end while
return -1

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2 - Time Complexity

Linear Search: Best: $\Omega(1)$ | worst: $O(n)$ | Average: $O(n)$

Binary Search: Best: $\Omega(1)$ | worst: $O(\log n)$ | Average: $O(\log n)$

3 - Comparison:

Binary Search	Linear Search	Feature
Must be Sorted	Any Arrang	Data requirement
$O(\log n)$	$O(n)$	Worst case Time
$O(1)$	$O(1)$	Space complexity

4 - Impact of Data Size and Arrangement:

Data Arrangement: Unsorted Data, Sorted Data

Input size (n): Small, large.